Curriculum Vitae

IVAN B. PENKOV

Personal

US citizen, born May 19, 1957, in Sofia, Bulgaria,
wife Irina Sousokolova, son Boyan and daughter Anna

Education

1979-82 — Undergraduate, Algebra, Dept. of Mathematics,
Moscow State University, Moscow, Russia
(Adviser: Professor Yu. I. Manin)

1983-87 — Ph.D. student, Dept. of Algebra Steklov
Mathematical Institute, Moscow, Russia
(Adviser: Professor Yu. I. Manin)

Degrees

June 1982 — Master Degree in Mathematics
Moscow State University
Thesis: The inverse Penrose transform
Adviser: Professor Yu. I. Manin

Oct. 1987 — Candidate of Physical and Mathematical Sciences (Ph.D.),
Steklov Mathematical Institute
Thesis: Typical modules over classical Lie superalgebras and their geometric realizations
Adviser: Professor Yu. I. Manin
Official referees: Professors D. Zhelobenko and A. Rudakov

Professional Experience

1982-83 — Research Assistant, Institute of Mathematics
Bulgarian Academy of Sciences, Sofia

May-June 1986 — Visiting Professor at Institute Fourier, Grenoble, France

1987-89 — Research Fellow, Section of Algebra, Institute of Mathematics, Bulgarian Academy of Sciences, Sofia, Bulgaria

Oct. 1989	Visiting Professor at the University of Paris VII, France
Jan.-Dec. 1990	Visiting Assistant Professor, Department of Mathematics, University of California at Berkeley
June-July 1990	Visiting Scholar at the University of Göttingen, Germany
Jan.–June 1991	Visiting Assistant Professor, Department of Mathematics, University of California at Davis
July 1991–June 1993	Assistant Professor, Department of Mathematics, University of California at Riverside
Sept.–Oct. 1993	Visiting Scholar at the Erwin Schrödinger Institute, Vienna, Austria
July 1993–June 2000	Associate Professor (tenured appointment), Department of Mathematics, University of California at Riverside
Apr.–May 1996	Visiting Scholar and Program Organizer at The Erwin Schrödinger Institute, Vienna, Austria
June 1996	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
June 1997	Visiting Professor at the University of Burgundy Dijon, France
Sept. 1997	Visiting Professor at the University Louis Pasteur, Strasbourg, France
Aug.–Sept. 1998	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Nov. 1999	Visiting Professor at the University of Hamburg, Germany
April-June 2000	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
July 2000 – Oct 2004	Professor (tenured appointment), Department of Mathematics, University of California at Riverside

Aug.–Sept. 2001	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Jan.–March 2002	Program organizer and MSRI scholar, MSRI, Berkeley
Aug.–Sept. 2002	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Fall 2003	Visiting Professor at Yale University
Spring 2004	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Fall 2007	Visiting Professor at University of California at Berkeley
Summer 2008	Visiting Professor at the University of Sao Paulo, Brazil
Spring 2012	Visiting Professor at Yale University
Fall 2012	Visiting Scholar at the Max-Planck Institute of Mathematics, Bonn, Germany
Fall 2004–present	Professor, Jacobs University Bremen, Germany (named International University Bremen before 2007)

Fields of Expertise

Representations of finite and infinite dimensional Lie algebras, generalized Harish-Chandra modules, representations of Lie superalgebras and Lie supergroups, geometry of supermanifolds, homogeneous superspaces, super D-modules, homogeneous ind-spaces and geometry of ind-varieties.

Grants, Awards, Organization of Scientific Programs

1984	First prize for young research mathematicians at the 4th Balkan Mathematical Olympiad, Sofia, for work on holomorphic geometry and supergeometry.
1989-1990	Participant in the NSF research grant for Bulgaria.

1991-present	Founding coorganizer of the California Lie Theory Program. This is a series of seminars at various UC campuses which meets 3 to 4 times a year.
1991-1995	NSF research grant (jointly with R. E. Block and V. Chari).
1995	NSF grant for organizing the conference in honor of R. E. Block (jointly with V. Chari).
1995	Grant from the Erwin Schrödinger Institute for the program "Representation Theory with Applications to Mathematical Physics" (jointly with Joseph A. Wolf).
1996	NSF grant for support of US graduate students at the Erwin Schrödinger Institute.
1995–1997	NSF research grant.
1997–2002	NSF Group Infrastructure Grant "Lie Groups, Lie Algebras and Their Representations" (jointly with Joseph A. Wolf and Geoffrey Mason).
Jan–May, 2002	Coorganizer of the one-semester MSRI Program "Representations of Infinite Dimensional Algebras and Mathematical Physics" (jointly with Edward Frenkel, Victor Kac, Vera Serganova and Gregg Zuckerman).
April 2002	Coorganizer of the MSRI Workshop "Quantum field theory and supersymmetry".
Sept. 2003	Coorganizer of the workshop "Locally finite Lie algebras at the Banff Research Institute, Canada.
2003–2006	NSF Grant "Lie Groups, Lie Algebras and Their Representations" (jointly with Joseph A. Wolf and Geoffrey Mason).

2004–present	Founding coorganizer of the Program "Darstellungstheorie, Transformationsgruppen und Mathematische Physik" of the consortium of five universities: IU-Bremen and the Universities of Bochum, Darmstadt, Hamburg, and Köln.
2005	Coorganizer of the International Conference in Honor of R.O. Wells, Jr., IUB, December 1-2, 2005.
2005	Grant for Research in Pairs, MF Oberwolfach.
2006	Grant for Research in Pairs, MF Oberwolfach, and for organizing a workshop at the International Research Station for Mathematical Innovation and Discovery at Banff(Canada).
2009–2012	DFG grant (one postdoc position).
2009	Volkswagen Foundation grant for organizing the summer school "Structures in Lie Representation Theory" (jointly with A. Joseph and A. Melnikov).
2009–2012	DFG grant within the priority program "SPP 1388 Darstellungstheorie"
2010	Coorganizer of the conference "Infinite-dimensional Lie algebras" at Newfoundland (jointly with Yu. Bahturin and V. Futorny).
2011	Coorganizer of the conference "Lie Groups" in honor of J.A. Wolf, Bochum, Germany
2012	Coorganizer of the German-Israeli workshop "Symplectic Geometry with applications to Representation Theory" at Jacobs University, Bremen, Germany
2012	Organizer of the meeting "7 1/2" in honor of Yu. I. Manin at IHES, Bures sur Yvette, France
2012–2013	DFG grant for German-Israeli cooperation

2013	Organzier of the meeting "Geometry" in honor of Vasil Tsanov, Sofia, Bulgaria
2012–2015	DFG grant within the priority program "SPP 1388 Darstellungstheorie"
2014	Coorganizer of the annual meeting in Soltau of the priority program "SPP 1388 Darstellungstheorie".
2014	Coorganizer of the meeting "Hodge theory and unitary representations of reductive Lie groups", Jacobs University Bremen.
2015	Coorganizer of the meeting "Groups and Rings Theory and Applications", July 15-22, Sofia, Bulgaria.
2015–2017	DFG grant (3/4 postdoc position)

Ph.D. Students

1994–1998	Ivan Dimitrov, 2001-2002 AMS Centennial Fellow, currently Associate Professor at Queen's University, Canada
1997–2001	Cyrill Oseledets.
1998–2003	Dimitar Grantcharov, 2003 UCR graduate research award, currently Associate Professor at the University of Texas Arlington.
2007–2010	Todor Milev, currently Lecturer at the University of Massachusetts, Boston.
2007–2010	Sergei Markouski
2008–2011	Alexey Petukhov, currently PostDoctoral Fellow at the University of Manchester.
2009–2013	Elitza Hristova, currently Researcher at the Bulgarian Academy of Sciences.
2014–	Thanasin Nampaisarn

Seminar Talks

1982–1987	About 10 talks at Moscow seminars run by Yu. I. Manin, A.A. Kirillov, I. M. Gelfand. Topics included: supergeometry, geometric methods in representation theory of Lie superalgebras, super D-modules.
1990	Several talks at Berkeley in the seminars of J. Wolf and M. Wodzicki. Invited lecture at the Northwest Geometry Seminar in Corvallis (Oregon). Invited talks in R. Howe's seminar at Yale University and in E. Binz's seminar at the University of Mannheim (Germany).
1991	Talk at the Berkeley meeting of the California Lie Theory Program.
1992	Talk at the Lie Groups and Lie Algebras Conference at UC San Diego. Talk at the Davis meeting of the California Lie Theory Program. Invited talk at the joint UCLA-UCR algebraic geometry seminar.
1993	Talk at the UCSC meeting of the California Lie Theory Program. Invited talks at the University of Graz (Austria), the University of Mannheim (Germany), Yale University, the Institute of Mathematics of the Bulgarian Academy of Sciences, Sofia (Bulgaria), the Max-Planck Institute of Mathematics, Bonn (Germany), and at Queen's University, Kingston (Canada).
1994	Invited talks at the Mathematical Physics Seminar, University of Southern California, Los Angeles, at Oklahoma State University and at the Institute of Mathematics of the Bulgarian Academy of Sciences.

1995	Invited talks at Texas A&M University, College Station (Texas), Yale University, MIT (V.Kac's seminar), University of Pennsylvania (A.A.Kirillov's seminar), the University of California at Santa Barbara, the Mathematical Institute of the Russian Academy of Sciences in St. Petersburg, Sofia University (Bulgaria),and at the National Technical University in Athens (Greece). Talk at the Berkeley meeting of the California Lie theory Program.
1996	Invited talks at the Fourier Institute in Grenoble, the University of Burgundy in Dijon, the University of Lyon, the University Paris VII and at the Max-Planck Institute for Mathematics in Bonn.
1997	Talks at both UCLA meetings of the California Lie Theory Program. Invited talks at Ecole Normale Superieure, Lyon, Laboratory for Mathematical Physics of the Bulgarian Academy of Sciences, Sofia (Bulgaria), the University Louis Pasteur, Strasbourg, the Fourier Institute in Grenoble, and the University of Wisconsin, Madison.
1998	Invited talks at the Representation theory seminars at Yale University and Rutgers University, at the Max Planck Institute in Bonn, Germany, and at the Algebra Seminar at UC Irvine.
1999	Invited speaker at the Representation Theory Seminar at Yale University.
2000	Invited talk at the Algebra Seminar at UC Irvine. Two talks at the Max-Planck Institute for Mathematics in Bonn (at the Ober-seminar and at Yu. I. Manin's seminar). Invited talk at University Lyon 1 (France). Invited talk at the Representation Theory Seminar at Yale University.

2001	Invited talk at the UCLA meeting of the California Lie Theory Program. Talk at the Berkeley miniworkshop of the California Lie Theory Program. Invited talks at Sofia University (V. Tsanov's seminar) and at The Bulgarian Academy of Sciences (I. Todorov's seminar). Talk at Yu. I. Manin's seminar at Max-Plank Institute of Mathematics in Bonn.
2002	Talk at MSRI within the program "Representations of Finite Dimensional Algebras and Mathematical Physics". Invited talk at I. Todorov's seminar at the Bulgarian Academy of Sciences.
2003	Invited talk at the Algebra seminar at the University of Connecticut. Invited talk at the Lie Groups seminar in Cornell University.
2004	Talks at Yu.I. Manin's seminar and the Oberseminar at the Max-Planck Institute for Mathematics in Bonn. Invited talks at SISSA, Trieste (Italy), at the International University Bremen, UC Santa Barbara, UC San Diego, UC Berkeley, and at the meeting of the inter-university seminar "Darstellungstheorie, Transformationsgruppen und Mathematische Physik" at the University of Bochum.
2005	Invited talk at the Eugene meeting of the California Lie Theory program.
2006	Invited talk at the seminar of M. Wodzicki at UCBerkeley. Invited talk at the Oberseminar at the University of Koeln.
2007	Invited talk at the Berkeley meeting of the California Lie Theory Program.
2008	Invited talk at the University of Aarhus, (Denmark). Invited talk at the University of Sao Paulo, (Brazil). Invited talk at the University of Campinas, Brazil.

2009	Invited talks at Yu.I. Manin's seminar at the Max-Planck Institute for Mathematics in Bonn, at the Emmy Noether Center in Erlangen, and at I. Todorov's seminar at the Bulgarian Academy of Sciences. Invited talk at Yale University on the occasion of G. Zuckerman's 60th birthday.
2010	Invited talks at Yu.I. Manin's seminar at the Max-Planck Institute for Mathematics in Bonn.
2011	Two invited talks at the University of Freiburg. Invited talk at Yu.I. Manin's seminar at the Max-Planck Institute for Mathematics in Bonn. Two talks at the Bremen-Hamburg-Kiel Seminar. Invited talk at the University of Zagreb, Croatia.
2012	Two invited talks at the Algebra and Lie Groups seminar at Yale University. Invited talks at the VIGRE seminar and at the Algebra seminar at the University of Georgia, Athens. Invited talk at the Tyurin Memorial meeting at the Max-Planck Institute for Mathematics in Bonn.
2013	Invited seminar talks at the universities of Jena and Münster. Invited talk at Yu.I. Manin's seminar at the Max-Planck Institute for Mathematics in Bonn.
2014	Invited seminar talk at the Weizmann Institute of Science, Israel.
2015	Invited seminar talks at UCSD, USC, UC Berkeley, USA. Invited talk at Yu.I. Manin's seminar at the Max-Planck Institute for Mathematics in Bonn.

Invited Series of Research Lectures

1983	Series of 3 invited lectures during the semester of mathematical physics in the Banach Institute, Warsaw.
1986	Series of 6 invited lectures in J.-L.Koszul's seminar at the Fourier Institute, Grenoble.
1988	Series of 3 invited lectures at the School in Geometry & Physics, Srni, Czechoslovakia.
1989	Series of 3 invited lectures in M. Duflo's seminar, at Universite Paris VII.
1990	Series of 3 invited lectures in H. Flenner's seminar at the University of Göttingen.
1993	Series of 2 invited lectures at the Erwin Schrödinger Institute in Vienna.
1997	Series of 2 invited lectures at M. Flato's seminar at the University of Burgundy in Dijon.
1999	Series of 8 lectures at the University of Hamburg.
2008	Series of 4 invited lectures at the Weizmann Institute of Science, Israel.
2008	Series of 5 invited lectures at the Escola de Algebra, Rio de Janeiro, Brazil.
2010	Series of 3 invited lectures at the School in Geometry & Physics, Srni, Czech Republic.
2014	Series of 3 invited lectures at the winter school "Lie algebras, algebraic groups and invariant theory", Moscow, Russia.

Conference Talks

1982–1993	Invited speaker at several international conferences, in particular the conference in honor of J.-L. Koszul (Institute Fourier, Grenoble, 1987) and the XIX DGMTP meeting (Rapallo, Italy, 1990).
1994	Speaker at the AMS conference on representation theory in Manhattan, Kansas. Invited plenary speaker at the conference "Classical & Quantum Geometry of homogeneous spaces", Moscow.
1995	Invited speaker at the conference "Modular Interfaces", Riverside.
1996	Invited speaker at the program "Representation Theory with Applications to Mathematical Physics", Vienna.
1997	Invited speaker at the international conference in honor of Yu.I.Manin, at the University of Antwerp, Belgium. Invited speaker at the Hayasibara Forum, RIMS, Kyoto, Japan.
2001	Invited speaker at the Midwest Lie Theory Conference (UWM Milwaukee). Invited speaker at the Darmstadt Lie Theory Workshop, Darmstadt, Germany.
2002	Invited speaker at the Darmstadt Lie Theory Workshop, Darmstadt, Germany. Invited speaker at the conference "Representations of Lie Groups, Harmonic Analysis on Homogeneous Spaces and Quantization", Leiden, The Netherlands.
2003	Invited speaker at the workshop "Infinite Dimensional Lie Theory and Its Application", Fields Institute, Toronto, Canada. Invited speaker at the Banff International Research Station workshop "Locally Finite Lie Algebras", Canada.

2004	Invited speaker at the joint Bremen-Göttingen workshop at the University of Göttingen.
2005	Invited speaker at the conference in honor of Yu. I. Manin at the Max Planck Institute for Mathematics, Bonn.
	Invited talk of the International Conference in Honor of R.O. Wells, Jr., IUB, December 1-2, 2005.
2006	Invited speaker at the workshop in Representation Theory, Weizmann Institute, Israel.
2008	Invited speaker at the meeting "Groups, Rings, Group-Rings", Ubatuba, Brazil.
2010	Invited speaker at the conference "Representations of Lie Groups and Algebraic Groups", Erlangen (September 2010). Invited speaker at the California Lie Theory Program, Berkeley (October 2010). Invited speaker at the meeting "Infinite-dimensional Lie Theory", Oberwolfach (Novermber 2010).
2012	Invited talk at the conference "Lie algebras and applications", Uppsala University, Sweden.
2013	Invited talk at the "Spring School on orbits, primitive ideals and quantum groups", Weizmann Institute, Rehovot, Israel. Invited talk at the Oberwolfach workshop "Representations of Lie groups and supergroups". Invited talk of the annual meeting of the DFG priority program 1388 "Representation theory" in Bad Boll, Germany. Invited talk at the conference "Representation Theory XIII" in Dubrovnik.

2014	Invited speaker at the workshop "Twelfth Night" at the Weizmann Institute of Science, Israel. Invited seminar talk at the University "Henri Poincaré", Nancy, France. Invited seminar talk at the University of Cagliari, Italy.
2015	Invited talk at the "Representation Theory Workshop", 7-9 May, Uppsala, Sweden. Invited talk at the XI. International Workshop "Lie Theory and its Applications in Physics", 15-21 June, Varna, Bulgaria. Invited speaker at the 49th "Seminar Sophus Lie", 22-25 September, Erlangen, Germany

Colloquium Talks

1985	University of Patras (Greece)
1986	Ecole Normale Superieure, Paris (France)
1989	J.W. von Goethe University, Frankfurt (Germany)
1989	University of Kaiserslautern (Germany)
1989	Institute Fourier, Grenoble (France)
1989	University of Mannheim (Germany)
1990	Rice University, Houston
1990	University of California, Davis
1991	University of Illinois, Urbana
1991	University of Indiana-Purdue, Indianapolis
1991	University of California, Riverside
1991	Oklahoma State University, Stillwater
1991	University of Iowa, Iowa City
1991	Howard University, Washington D.C.
1991	University of Oregon, Eugene
1992	University of California, Santa Cruz
1992	University of Utah, Salt Lake City
1993	Erwin Schroedinger Institute, Vienna (Austria)
1993	J.W. von Goethe University, Frankfurt (Germany)
1993	Queen's University, Kingston (Canada)
1995	University of Southern California, Los Angeles
1997	University of California, Berkeley

1997	California Polytechnic State University, San Luis Obispo
1997	University of Wisconsin, Milwaukee
1998	University of California, Davis
1998	Clairmont Colleges, Clairmont
1999	University of Oklahoma, Norman (Karcher Colloquium)
1999	University of Hamburg (Germany)
2002	Oklahoma State University, Stillwater
2003	Yale University
2003	Queen's University, Kingston (Canada)
2005	University of Hamburg (Germany)
2006	University of Tübingen (Germany)
2006	University of Bremen (Germany)
2007	Yale University
2007	University of California at Santa Barbara
2007	California Polytechnic State University, San Luis Obispo
2007	University of San Jose
2011	University of Freiburg (Germany)
2012	Yale University
2014	Independent University in Moscow (Russia)
2015	University of Oldenburg (Germany)

List of Publications
IVAN B. PENKOV

Research papers

1. The Penrose transform on general grassmannians, C.R. Acad. Sci. Bulg., 33 (1980), 1439-1442.

2. Null-geodesics of complex Einstein spaces, Funk. Anal. i Priloz., 16, No.1 (1982), 78-79,(Russian), (with Yu.I.Manin).

3. Linear differential operators and cohomology of analytic spaces, Uspekhi Mat. Nauk, 37, No 4 (1982), 171-172, (Russian).

4. D-modules on supermanifolds, Invent.Math., 71 (1983), 501-512.

5. The formalism of left and right connections on supermanifolds, (1982), Lect. on supermanifolds, geom. methods and conf. groups, editor: Döbner et al, World Scientific, Singapore, 1989, 3-13 (with Yu. I. Manin).

6. An introduction to geometric representation theory for complex simple Lie superalgebras, Proc. Conf. Diff. Geom. Math. in Physics, Shumen 1984, World Scientific 1986, 89-106.

7. Serre duality for projective supermanifolds, Funk. Anal. i Priloz., 18,No.1 (1984), 78-79, (Russian) (with O. Ogievetsky).

8. Projectivity and D-affinity of flag supermanifolds, Uspekhi Math. Nauk, 40 No.1 (1985), 211-212, (Russian) (with I. Skornyakov).

9. A geometric approach to the linear Penrose transform, Transactions AMS, 290 (1985), 555-575.

10. Cohomologie des D-modules typiques sur les supervariétés de drapeaux, C. R. Acad. Sci., Paris, 199 (1984), 1005-1008 (with I. Skornyakov).

11. Characters of typical irreducible finite dimensional $q(m)$-modules, Funk. Anal. i Priloz., 20 (1986), No. 1, 37-45 (Russian).

12. Localisation des representations typiques d'une superalgèbre de Lie complexe classique, C. R. Acad. Sci. Paris, 304 (1987), 163-166.

13. Borel-Weil-Bott theory for classical Lie supergroups, Curr. Probl. Math., Moscow, 32 (1988), 71-124 (Russian).

14. Elements of supergeometry, Curr. Probl. Math., Moscow, 32 (1988), 3-25, (Russian), (with Yu. I. Manin and A. Voronov).

15. Character formulas for some classes of atypical $gl(m+n\varepsilon)$-and $p(m)$-modules, Lett. Math. Phys., 16 (1988), 251-261 (with V. Serganova).

16. Cohomology of G/P for classical complex Lie supergroups G and characters of some atypical G-modules, Annales Fourier, 39 (1989), 845-873, (with V. Serganova).

17. Generic irreducible representations of classical Lie superalgebras, Lecture Notes in Physics, 375 (1991), Springer Verlag, Berlin, 311-319 (with V. Serganova).

18. Representations of classical Lie superalgebras of type 1, Indag. Math., N.S. 3(2), (1992), 419-466 (with V. Serganova).

19. Generic representations of classical Lie superalgebras and their localisation, Monatsh. Math., 118 (1994), 267-313.

20. Generic irreducible representations of finite-dimensional Lie superalgebras, Intern. Journ. Math., 5 (1994), 389-419 (with V. Serganova).

21. Partially and fully integrable modules over Lie superalgebras, Studies in Advanced Mathematics (series editor S.-T. Yau), 4, AMS and Internatl. Press 1997, 49-67 (with I. Dimitrov).

22. Characters of irreducible G-modules and cohomology of G/P for the Lie supergroup $G = Q(n)$, Journ. of Math. Sci. 84 (1997), 1382-1412 (with V. Serganova).

23. Characters of finite-dimensional irreducible $q(n)$-modules, Lett. Math. Phys. 40 (1997), 147-158 (with V. Serganova).

24. Characters of strongly generic irreducible Lie superalgebra representations, Intern. Journ. Math. 9 (1998), 331-366.

25. Partially integrable highest weight modules, Transformation Groups 3 (1998), 241-253 (with I. Dimitrov).

26. The support of an irreducible Lie algebra representation, Journ. of Algebra 209 (1998), 129-142 (with V. Serganova).

27. Weight modules of direct limit Lie algebras, IMRN 1999, no. 5, 223-249 (with I. Dimitrov).

28. Weight representations of the Cartan type Lie algebras W_n and $\bar{S}_n$, Math. Research Lett. 6 (1999), 397-416. (with V. Serganova).

29. On the structure of weight modules, Transactions AMS 352 (2000), pp. 2857-2869 (with I. Dimitrov and O. Mathieu).

30. Generalized Harish-Chandra modules, Moscow Math. Journal 2 (2002), 753-767 (with V. Serganova).

31. A Bott-Borel-Weil theory for direct limits of algebraic groups, American Journ. of Math. 124 (2002), 955-998 (with I. Dimitrov and J. A. Wolf).

32. Locally finite Lie algebras with root decomposition, Archiv der Math. 80 (2003) 478-485 (with H. Strade).

33. Cartan subalgebras of gl_∞, Canadian Math. Bull., 46 (4), 2003, 597-616 (with K.-H. Neeb).

34. Finite rank vector bundles on inductive limits of grassmannians, IMRN 2003, no. 34, 1871-1887 (with J. Donin).

35. Classically semisimple locally finite Lie superalgebras, Forum Math. 16 (2004), 431-446.

36. On the existence of (g,k)-modules of finite type, Duke Math. Journal 125 (2004), 329-349 (with V. Serganova and G. Zuckerman).

37. A Reduction Theorem for Highest Weight Modules over Toroidal Lie Algebras, Comm. Math. Physics 250(2004), no. 1, 47-53 (with I. Dimitrov and V. Futorny).

38. Generalized Harish-Chandra modules: a new direction of the structure theory of representations, Acta Applicandae Mathematicae, 81(2004), 311-326(with G. Zuckerman).

39. Ind-varieties of generalized flags as homogeneous spaces for classical ind-groups, IMRN 2004, no. 55, 2935-2953(with I. Dimitrov).

40. Borel subalgebras of $gl(\infty)$, Resenhas IME-USP 6 (2004), 111-119(with I. Dimitrov).

41. Generalized Harish-Chandra modules with generic minimal k-type, Asian

Journal of Mathematics 8(2004), 795-812(with G. Zuckerman).

42. A construction of generalized Harish-Chandra modules with arbitrary minimal k-type, Canad. Math. Bull. 50 (2007), 603-609 (with G. Zuckerman).

43. Cartan subalgebras of root-reductive Lie algebras, Journal of Algebra, 308 (2007), 583-611 (with E. Dan-Cohen and N. Snyder).

44. A construction of generalized Harish-Chandra modules for locally reductive Lie Algebras, Transformation Groups 13 (2008), 799-817 (with G. Zuckerman).

45. Locally semisimple and maximal subalgebras of the finitary Lie algebras $gl(\infty)$, $sl(\infty)$, $so(\infty)$, and $sp(\infty)$, Journal of Algebra 322 (2009), 2069-2081 (with I. Dimitrov).

46. Parabolic and Levi subalgebras of finitary Lie algebras, IMRN, International Mathematics Research Notices, Article ID rnp169, doi:10.1093/imrn/rnp169 (with E. Dan-Cohen).

47. Parabolic Subgroups of Real Direct Limit Lie Groups, Contemporary Mathematics 499, 2009, pp. 47-59 (with E. Dan-Cohen and J. A. Wolf).

48. Rank 2 vector bundles on ind-grassmannians, Manin Festschrift,in Progress in Mathematics 270, Birkhaeuser Boston, 2009, pp. 555-572 (with A. S. Tikhomirov).

49. Bounded simple (g; sl(2))-modules for rkg = 2, Journal of Lie Theory 20 (2010), 581-615 (with V. Serganova).

50. Tensor representations of infinite-dimensional root-reductive Lie algebras, in Developments and Trends in Infinite-Dimensional Lie Theory, Progress in Mathematics 288, Birkhäuser, 2011, pp. 127-150 (with K. Styrkas).

51. On bounded generalized Harish-Chandra modules, Annales de l'Institut Fourier, 62 (2012), 477-496 (with V. Serganova).

52. Triviality of vector bundles on twisted ind-Grassmannians, Mat. Sbornik 202:1 (2011), 1-39 (with A. S. Tikhomirov).

53. A Bott-Borel-Weil theorem for diagonal ind-groups, Canadian Journal of Mathematics, doi:10.4153/CJM-2011-032-6 (with I. Dimitrov).

54. Categories of integrable $sl(\infty)$-, $o(\infty)$-, $sp(\infty)$-modules, in "Representation Theory and Mathematical Physics", Contemporary Mathematics 557, AMS 2011, pp. 335-357 (with V. Serganova).

55. Levi components of parabolic subalgebras of finitary Lie algebras, in "Representation Theory and Mathematical Physics", Contemporary Mathematics 557, AMS 2011, pp. 129-149 (with E. Dan-Cohen).

56. A Koszul category of representations of finitary Lie algebras, Advances in Mathematics (to appear), arXiv:1105.3407 (with E. Dan-Cohen and V. Serganova).

57. On the structure of the fundamental series of generalized Harish-Chandra modules, Asian Journal of Mathematics , 16(2012), 489-514 (with G. Zuckerman).

58. On Ideals in the enveloping algebra of a locally simple Lie algebra, IMRN (to appear), arXiv:1210.0466 (with Alexey Petukhov).

59. Linear Ind-Grassmannians, Pure and Applied Math. Quarterly (to appear), arXiv:1310.8054 (with Alexander Tikhomirov).

60. Algebraic methods in the theory of generalized Harish-Chandra modules, Developments and Retrospectives in Lie Theory: Algebraic Methods, Springer Verlag (to appear), arXiv:1310.8058 (with Gregg Zuckerman).

61. Tensor representations of Mackey Lie algebras and their dense subalgebras, Developments and Retrospectives in Lie Theory: Algebraic Methods, Springer Verlag (to appear), arXiv:1311.5217 (with Vera Serganova).

62. On the Barth-Van de Ven-Tyurin-Sato Theorem, preprint 2014, arXiv:1405.3897 (with Alexander Tikhomirov).

63. A categorification of the Boson-Fermion correspondence via Representation Theory of $sl(\infty)$, preprint 2014, arXiv:1410.7553 (with Igor Frenkel and Vera

Serganova).

64. Annihilators of highest weight $sl(\infty)$-modules, preprint 2014, arXiv:1410.8430 (with Alexey Petukhov).

65. Infinite Konstant cascades and centrally generated primitive ideals of $U(n)$ in types A_∞, C_∞, preprint 2015, arXiv:1502.05486 (with Mikhail Ignatyev).

66. Schubert decompositions for ind-varieties of generalized flags, arXiv:1506.08263, preprint 2015 (with L. Fresse).

67. On ideals in $U(sl(\infty))$, $U(o(\infty))$, $U(sp(\infty))$, arXiv:1511.01733, preprint 2015 (with A. Petukhov).

Books edited

1. V. Chari, I. B. Penkov - editors. Modular Interfaces: Modular Lie Algebras, Quantum Groups, and Lie Superalgebras. "Studies in Advanced Mathematics", vol. 4, 1997, AMS.

2. A. Joseph, A. Melnikov, I. Penkov - editors. Highlights in Lie Algebraic Methods. "Progress in Mathematics", vol. 295, 2011, Birkhauser.

3. A. Huckleberry, I. Penkov, G. Zuckerman - editors. Lie Groups: Structure, Actions, and Representations. "Progress in Mathematics", vol. 306, 2013, Birkhauser.